cuemath

Mensuration and Solids

Review Exercise Questions

Level-2

Miscellaneous

Q1. A pipe with a rectangular cross-section is used to fill up a tank. It takes 36 hours to fill the tank completely. Now, the cross-sectional dimensions of the pipe are increased by 50%, but the flow rate is decreased by 50%. In how much time will the tank be filled completely in the second situation?

Q2. Two rectangular pipes A and B with dimensions 3 cm × 4 cm and 6 cm × 3 cm respectively feed into a cylindrical tank with base radius 1.2 m. The flow rate of water through both the pipes (when opened) is 10 cm/sec. Pipe A is opened for 10 minutes and then closed. Next, Pipe B is opened for 25 minutes and then closed. What will be the height of the water in the cylindrical tank now?

Q3. 100 identical pipes of cross-sectional dimensions 10 cm × 8 cm feed into an underground hemispherical reservoir of radius 25 m. A system maintains the flow rate of water through each pipe at 20 cm/sec. Another control system opens the pipes for 2 hours every night and keeps them closed for the remaining time of the day. There is a field above the reservoir, of dimensions 300 m × 150 m. The region around the field receives rainfall of 4 cm during the daytime every day in the rainy season. A mechanism is designed to collect the rainwater falling on the field and channel it into the reservoir through the system of pipes.

A. How much water is wasted every day because of this flow control?
B. How many days of rainfall is required to fill the reservoir in this situation?
C. To what minimum value should the flow rate be increased so that all the rainwater on a given day is harvested?
D. How much time would it take to fill the reservoir in the case of this increased flow rate?

Q4. There are 3 rooms in a house. The details of each room are provided in the table below:

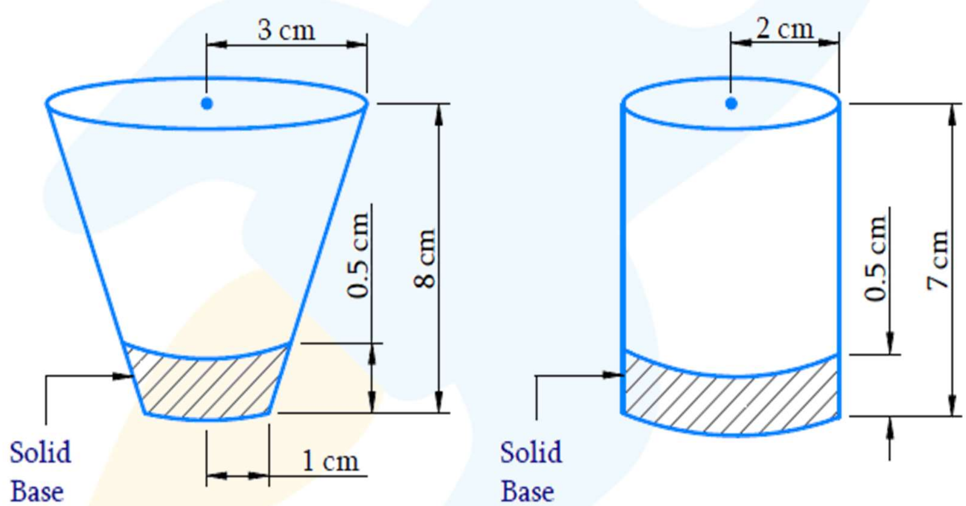
	Inner dimensions	**No. of doors**	**No. of windows**
Room 1	10 m × 8 m × 6 m	2	4
Room 2	12 m × 8 m × 6 m	2	4
Room 3	8 m × 6 m × 6 m	1	2

The walls of Rooms 1 and 2 will be coated with premium paint, while the walls of Room 3 will be coated with a low-grade paint. 1 liter of premium paint (which costs INR 2500/liter) can cover 12 square meters of wall, while 1 liter of low-grade paint (which costs INR 1500/liter) can cover 16 square meters of wall. The dimensions of each door are 2 m × 3 m, while the dimensions of each window are 1 m × 2 m.

What will be the cost involved in this paint job in lacs (round off your answer to two decimal places)? Note that the walls include the ceilings as well, but not the floors.

Q5. A smuggler is trying to smuggle a highly valuable liquid across a border. The cost of this liquid in the international market is INR 5 crores/liter. The smuggler designs an ingenious way to carry out this task. He designs a closed cuboidal container of outer dimensions 50 cm × 27 cm × 20 cm, with the walls, the base and the lid of thickness 1 cm. Inside, he places two equally spaced partitions along the length, each of thickness 1.5 cm, thus dividing the container into three compartments. He then goes on to divide each compartment with another partition of thickness 2 cm (along the height), thus dividing each compartment into two equal sub-compartments. In the bottom sub-compartments, he stores the valuable liquid, while the top sub-compartments are used to store dry-fruits, to fool the customs officials. How much money (in crores) will he make if he succeeds in his task and is able to sell the liquid?

Q6. A tea-seller has a hemispherical vessel, in which he stores his tea. The outer diameter of this vessel is 32 cm, and the thickness of its wall is 1 cm. The tea-seller has two kinds of glasses, whose geometrical structures are shown below:



One day, he uses the cylindrical glasses to sell tea, while the next day, he uses the other glasses (the other shape is known as a *frustum* of a cone). If each day, he starts selling tea with a full vessel, sells a glass at Rs. 6, and finishes selling the entire tea, what is the difference in the money made on the two days?